535/1
PHYSICS
Paper 1
July /August
2½ Hours

AITEL JOINT MOCK ASSESSMENTS 2025

Uganda Certificate Of Education

PHYSICS

Paper 1

(Theory)

2 Hours 30 minutes

INSTRUCTIONS TO CANDIDATES.

This paper consists of two sections; A and B It has seven examination items..

Section A has three compulsory items.

Section B has two parts; I and II. Answer one item from each part.

Answer five items in all.

Any additional item(s) answered will not be scored.

All answers must be written in the booklets provided.

©2025 AITEL JOINT MOCKS

SECTION A
Answer all the items from this section

ITEM: 1

A brass band was invited to play during a candidates' prom party near a tall building, a distance slightly more than 17 m away. Two friends standing at a distance from the playing band, heard the sound from the band which attracted them to go and attend the celebration.

On arrival, they were allowed to enter the main hall where celebrations were to be held where they stood between two walls of the main hall but 400m from the nearest. As the DJ was checking the speaker which was placed at point where friends were standing, they heard two returning sounds from the walls at different intervals which made them confused and decided to determine length of main hall. Later when the whole system was switched on the sound was very loud seemed destructive to ears, unclear, confused and indistinct. As the friends were moving around observing decorations, they observed scaring image of them in a curved shinny wall and on inquiring they were told that it's a mirror and can be used for several purposes Later in the night during the cerebration, colored lights flashing red, blue and green made the colors of their clothes look different from the original colors which puzzled them. The friends heard returning sounds, first after 2 s and the second 3 s later, respectively and were originally wearing yellow clothes.

Speed of sound in air = 330 ms^{-1}

Friends are of height 60 cm and standing 10 cm from curved surface of curvature 30 cm

Speakers produced sound waves of wave length 10mm

Task:

As a physics student, help the two friends to understand why;

(a) They heard different returning sounds

(b) If the sound produced by speakers is safe for learners' ears.

(c) the color of their clothes kept changing when colored lights flashed on them.

(d) They saw scaring image in the curved wall surface and how it useful in daily life

ITEM: 2

On a certain day at around 1:30pm while in the dining hall, learners watched heavy fog-rains and floods being experienced in a certain outside country on an international T.V live channel.

To worsen matters, the floods were happening at night and this risked many natives as many of them were ambushed while asleep. Learners wondered how it would be night and seriously raining in an area yet it was day and the Sun was highly shinning at that time in their school.

Task:

As a physics learner help the learners clear their queries about;

a) Occurrence of the floods in one area yet it was shining in their school at same time.

b) Why it was night in that outside country yet it was day-time in their area?

c) How T.V signals broadcast from where the floods were happening reached them.

©2025 AITEL JOINT MOCKS

ITEM: 3

A group of students visited the nuclear power plant and they were blocked from entrance due to an accident that had occurred and several operators were exposed to radiation. Only emergency cars were allowed to enter to transport the operators to the hospital to undergo medical examination, including x-ray scans to assess any potential radiation related injuries. Students wondered how harmful radiations were. It became hard for them to understand even how the power plant could generate electricity after being told at the entrance that the plant was the only source of electricity in that area. If the plant' s radiation monitoring system detects a background radiation level of 30 counts per minute, and when the radioactive source is placed in front of the radiation detector, the reading on the detector is 550 counts per minute and the source has a half-life of 25 minutes.

Hint: The power plant will be safe for use if the detector reads 32.5 counts per minute.

Task

As a student of physics;

(a) Advise the Power plant team leader on how long they had to wait for the power plant to be safe for use.

(b) Help the students to understand;
   (i) the relevant nuclear process at the power plant
   (ii) how x rays are produced and how they can identify an injury

(c) Assist learners to appreciate that indeed the guide was right to stop them and explain how the operators managed to be there always.

SECTION B

PART 1

Answer one item from this part

ITEM 4

A pump is used to fill a tank by drawing water from underground well to be used at school power house. In order to minimize on power consumption its use for limited time, the pump works at a constant rate of $2.4x10^2W$ and it's able to raise 0.188 m^3 of water every minute through a height of 6m. The cooks complain that it is slow at work. At the power house the cook is always surprised because water pumped in the morning is always above the normal temperature at 30°C when measured using a thermometer which the cook checks before making morning tea. The pump is said to be efficient if the ratio of its work out put to work input is above 0.75 and the cook boils 9.5l of water for morning tea and 500g evaporated.

Support materials

Density of water is $1000kgm^{-3}$

Specific heat capacity of water = $4200Jkg^{-1}K^{-1}$

Specific latent heat of vaporization of water $2.56 \times 106Jkg - ^1$

Task.

(a) Obtain the rate at which the pump works in raising the water and comment on its efficiency.

(b) Explain to the cooks why water pumped in the morning has such temperature.

(c) How much heat is necessary to turn the lost volume of water into vapor.

©2025 AITEL JOINT MOCKS

ITEM 5

On a construction site a builder uses an electric lifter to transfer the mortar from the ground to a height of 17.0m on the building. When power went off and work has to continue, the builders got a rope, a pail of negligible mass with a handle that lift a maximum of 20kg of mortar. They want to design a simple machine with a help of a grooved rim which can enable them resume work.
Given that the rope has a maximum thermal strength of 3.8kJ.
Hint;
Assume all the potential energy is converted to heat energy
Task.

(a) Help the builders to design the machine and guide the on how it works.
(b) Advise whether the rope will withstand the heat generated or it will break.
(c) Suggest the necessary factor to consider that enable the rope to success fully execute the work.

PART II

Answer one item from this part.

ITEM 6

A land-lady was recently told that electricity from a power substation is transmitted at 12 kV using thick aluminum cables and is to be used in her house at 240 V. She got confused by how the electricity changes voltage from one value to another and said that the thick cables are a waste of money. Inside the house are two sets of appliances of resistance 20 Ω and 30 Ω but the land-lady is not sure of how they should be connected to ensure that they will work effectively.
Use your knowledge of physics to help the land-lady to understand;

a) The system through how the electricity changes from one value to another.
b) Why such cables are used.
(d) How the appliances should be connected to ensure that they work effectively with a high amount of current

ITEM 7

A parent had a small piece of land besides his main house. He decided to put up a two roomed rental for purpose of increasing on his average income. He however set the rentals directly below electric transmission power lines and installed one electric meter to be used by two different occupants of his rentals. Misunderstandings arose as it was not clear which occupant to pay how much of electric bills. It was however discovered that the two occupants used one inside bulb each of different ratings and one common security bulb, during weekends they could use a generator but were worried about its mode of working claiming that it may be consuming electricity.
It was further discovered that ;
Occupant of room 1 was using a flat iron rated 1000W for 30 minutes per day, TV rated 150 W for four hours per day and a light bulb rated 7 W for 5 hours per day Occupant of room 2 was using a kettle rated 1500W for 40 minutes each day, mini refrigerator rated 500 W for 2 hours each day and a light bulb rated 7 W for 8 hours each day. The security light is rated 10 W for 5 hours each day
Hint : *cost of one unit is UGX 704*
The landlord tasked the engineer to prepare a report on the following

(i) If it is safe for the occupants to be in rooms below electric transmission lines
(ii) The amount that must be paid by each occupant in 30 days
(iii) What occupants should do to reduce consumption of electric power
(iv) Mode of operation of a generator

Task

Help the engineer to write the report

END

©2025 AITEL JOINT MOCKS